ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. IT ENGINEERING — SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 State Master's Theorem for recurrences. Explain Strassen's matrix multiplication. [14 Marks]

[OR]

Q.1 (Alt) Explain Amortized Analysis and Randomized Quick Sort with expected running time. [14 Marks]

--- UNIT II ---

Q.2 Solve Fractional Knapsack using Greedy approach. Explain Huffman Coding prefix property. [14 Marks]

[OR]

Q.2 (Alt) Explain Job Sequencing with Deadlines and Minimum Spanning Trees (Kruskal, Prim). [14 Marks]

--- UNIT III ---

Q.3 Solve 0/1 Knapsack using Dynamic Programming. Find Longest Common Subsequence (LCS). [14 Marks]

[OR]

Q.3 (Alt) Explain Matrix Chain Multiplication algorithm and optimal parenthesization. [14 Marks]

--- UNIT IV ---

Q.4 Explain Backtracking: Solve 8-Queens and Subset Sum problems. [14 Marks]

[OR]

Q.4 (Alt) Explain Branch and Bound: Solve 0/1 Knapsack and TSP with reduced cost matrices. [14 Marks]

--- UNIT V ---

Q.5 Define Complexity Classes: P, NP, NP-Hard, NP-Complete. State Cook's Theorem. [14 Marks]

[OR]

Q.5 (Alt) Prove 3-SAT and Vertex Cover problems are NP-Complete by polynomial reduction. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****